

Network Troubleshooting Lab 004

Default Gateway Troubleshooting

Scenario:

A user reports that their workstation is connected to the network but cannot access external resources such as websites or cloud services. The objective of this troubleshooting lab is to verify default gateway configuration, test routing behavior, and determine whether traffic can successfully leave the local network.

Document Information

Category: Networking / IT Support

Lab Type: Troubleshooting Case

Environment: MacBook Pro - macOS

Technician: Ridley Lherisse

Date: 10/05/2026

1. Troubleshooting Methodology

The troubleshooting process followed a structured approach:

1. Identify the reported issue
2. Verify network configuration
3. Test local network communication
4. Analyze routing information
5. Confirm external connectivity
6. Document findings and resolution

2. Step 001 — Review Routing Table

Command Used

netstat -rn

Purpose

Used to review the device's routing table and identify the default gateway responsible for forwarding traffic outside the local network.

Information Reviewed

- Default route
- Gateway address
- Network interfaces
- Routing entries

Evidence

```
fe80::893:c998:d497:67c0%en0      30:3b:7c:8b:f6:f9      UHLWI      en0
fe80::c50:6e65:ee37:a960%en0      72:b1:56:64:cb:a9      UHLWI      en0
fe80::c0f:8a5c:2819:33dd%en0      7e:7b:6a:bb:19        UHLWI      en0
fe80::105d:ca95:b238:31%en0        30:3b:7c:8b:f6:f9      UHLWI      en0
fe80::1286:8cff:fe7e:f841%en0      10:86:8c:7e:f8:41      UHLWIir    en0
fe80::143c:8c4f:8b5:8559%en0      86:ea:8b:31:56:79      UHLWIi     en0
fe80::107b:e32a:845c:a74%en0      1e:08:f2:1e:b8:6        UHLWI      en0
fe80::1cad:84e2:16c8:d323%en0      3c:6:30:33:b6:6d       UHLI       lo0
fe80::d079:0fff:fe44:2428%awd10    d2:79:6f:44:24:28      UHLI       lo0
fe80::d079:0fff:fe44:2428%llw0     d2:79:6f:44:24:28      UHLI       lo0
fe80::Nutun0/64                    fe80::af27:39e9:de17:1645%Nutun0 UCI         utun0
fe80::af27:39e9:de17:1645%Nutun0  link#15                UHLI       lo0
fe80::Nutun1/64                    fe80::5616:4fc:47eb:f6c%Nutun1  UCI         utun1
fe80::5616:4fc:47eb:f6c%Nutun1    link#16                UHLI       lo0
fe80::Nutun2/64                    fe80::ce81:b1c:bd2c:69e%Nutun2  UCI         utun2
fe80::ce81:b1c:bd2c:69e%Nutun2    link#17                UHLI       lo0
fe80::Nutun3/64                    fe80::f377:9484:e35a:ca86%Nutun3 UCI         utun3
fe80::f377:9484:e35a:ca86%Nutun3  link#18                UHLI       lo0
fe80::Nutun4/64                    fe80::c682:e93b:7892:55c3%Nutun4 UCI         utun4
fe80::c682:e93b:7892:55c3%Nutun4  link#19                UHLI       lo0
fe80::Nutun5/64                    fe80::8853:5a51:cfcc:f8c9%Nutun5 UCI         utun5
fe80::8853:5a51:cfcc:f8c9%Nutun5  link#20                UHLI       lo0
fe80::Nutun6/64                    fe80::af27:8863:c81:acbb%Nutun6  UCI         utun6
fe80::af27:8863:c81:acbb%Nutun6  link#21                UHLI       lo0
ff00::/8                           :1                      UmcI       lo0
ff00::/8                           link#4                  UmcI       anp10
ff00::/8                           link#5                  UmcI       anp11
ff00::/8                           link#11                 UmcI       ap1
ff00::/8                           link#12                 UmcI       en0
ff00::/8                           link#13                 UmcI       awd10
ff00::/8                           link#14                 UmcI       llw0
ff00::/8                           fe80::af27:39e9:de17:1645%Nutun0 UmcI       utun0
ff00::/8                           fe80::5616:4fc:47eb:f6c%Nutun1  UmcI       utun1
ff00::/8                           fe80::ce81:b1c:bd2c:69e%Nutun2  UmcI       utun2
ff00::/8                           fe80::f377:9484:e35a:ca86%Nutun3 UmcI       utun3
ff00::/8                           fe80::c682:e93b:7892:55c3%Nutun4 UmcI       utun4
ff00::/8                           fe80::8853:5a51:cfcc:f8c9%Nutun5 UmcI       utun5
ff00::/8                           fe80::af27:8863:c81:acbb%Nutun6  UmcI       utun6
ff01::%lo0/32                      :1                      UmcI       lo0
ff01::%Namp10/32                   link#4                  UmcI       anp10
ff01::%Namp11/32                   link#5                  UmcI       anp11
ff01::%Nap1/32                     link#11                 UmcI       ap1
ff01::%Nen0/32                     link#12                 UmcI       en0
ff01::Nutun0/32                    fe80::af27:39e9:de17:1645%Nutun0 UmcI       utun0
ff01::Nutun1/32                    fe80::5616:4fc:47eb:f6c%Nutun1  UmcI       utun1
ff01::Nutun2/32                    fe80::ce81:b1c:bd2c:69e%Nutun2  UmcI       utun2
ff01::Nutun3/32                    fe80::f377:9484:e35a:ca86%Nutun3 UmcI       utun3
ff01::Nutun4/32                    fe80::c682:e93b:7892:55c3%Nutun4 UmcI       utun4
ff01::Nutun5/32                    fe80::8853:5a51:cfcc:f8c9%Nutun5 UmcI       utun5
ff01::Nutun6/32                    fe80::af27:8863:c81:acbb%Nutun6  UmcI       utun6
ff02::%lo0/32                      :1                      UmcI       lo0
ff02::%Namp10/32                   link#4                  UmcI       anp10
ff02::%Namp11/32                   link#5                  UmcI       anp11
ff02::%Nap1/32                     link#11                 UmcI       ap1
ff02::%Nen0/32                     link#12                 UmcI       en0
ff02::Nutun0/32                    fe80::af27:39e9:de17:1645%Nutun0 UmcI       utun0
ff02::Nutun1/32                    fe80::5616:4fc:47eb:f6c%Nutun1  UmcI       utun1
ff02::Nutun2/32                    fe80::ce81:b1c:bd2c:69e%Nutun2  UmcI       utun2
ff02::Nutun3/32                    fe80::f377:9484:e35a:ca86%Nutun3 UmcI       utun3
ff02::Nutun4/32                    fe80::c682:e93b:7892:55c3%Nutun4 UmcI       utun4
ff02::Nutun5/32                    fe80::8853:5a51:cfcc:f8c9%Nutun5 UmcI       utun5
ff02::Nutun6/32                    fe80::af27:8863:c81:acbb%Nutun6  UmcI       utun6
```

ridleyherisse@Ridleys-MacBook-Pro ~ %

Findings

The routing table was reviewed to identify the default gateway configured on the device. The default route determines where traffic is sent when the destination is outside the local network.

3. Step 002 — Verify Default Gateway Connectivity

Command Used

ping 10.0.0.1

Purpose

Used to confirm communication between the workstation and the default gateway.

Evidence

```
fe80::Nutun3/64      fe80::f377:9484:e35a:ca86%utun3  UoI      utun3
fe80::f377:9484:e35a:ca86%utun3  link#18      UHlI      lo0
fe80::Nutun4/64      fe80::c682:e93b:7892:55c3%utun4  UoI      utun4
fe80::c682:e93b:7892:55c3%utun4  link#19      UHlI      lo0
fe80::Nutun5/64      fe80::8853:5a51:cfcc:f8c9%utun5  UoI      utun5
fe80::8853:5a51:cfcc:f8c9%utun5  link#20      UHlI      lo0
fe80::Nutun6/64      fe80::af27:8863:c81:acbb%utun6  UoI      utun6
fe80::af27:8863:c81:acbb%utun6  link#21      UHlI      lo0
ff00::/8             :1              UmCI      lo0
ff00::/8             link#4          UmCI      anp10
ff00::/8             link#5          UmCI      anp11
ff00::/8             link#11         UmCI      ep1
ff00::/8             link#12         UmCI      en0
ff00::/8             link#13         UmCI      awd10
ff00::/8             link#14         UmCI      llw0
ff00::/8             fe80::af27:39e9:de17:1645%utun0  UmCI      utun0
ff00::/8             fe80::5616:4fc4:7eb1:f6c%utun1  UmCI      utun1
ff00::/8             fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff00::/8             fe80::f377:9484:e35a:ca86%utun3  UmCI      utun3
ff00::/8             fe80::c682:e93b:7892:55c3%utun4  UmCI      utun4
ff00::/8             fe80::8853:5a51:cfcc:f8c9%utun5  UmCI      utun5
ff00::/8             fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ff01::%lo0/32        :1              UmCI      lo0
ff01::%anp10/32      link#4          UmCI      anp10
ff01::%anp11/32      link#5          UmCI      anp11
ff01::%ep1/32        link#11         UmCI      ep1
ff01::%en0/32        link#12         UmCI      en0
ff01::%utun0/32      fe80::af27:39e9:de17:1645%utun0  UmCI      utun0
ff01::%utun1/32      fe80::5616:4fc4:7eb1:f6c%utun1  UmCI      utun1
ff01::%utun2/32      fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff01::%utun3/32      fe80::f377:9484:e35a:ca86%utun3  UmCI      utun3
ff01::%utun4/32      fe80::c682:e93b:7892:55c3%utun4  UmCI      utun4
ff01::%utun5/32      fe80::8853:5a51:cfcc:f8c9%utun5  UmCI      utun5
ff01::%utun6/32      fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ff02::%lo0/32        :1              UmCI      lo0
ff02::%anp10/32      link#4          UmCI      anp10
ff02::%anp11/32      link#5          UmCI      anp11
ff02::%ep1/32        link#11         UmCI      ep1
ff02::%en0/32        link#12         UmCI      en0
ff02::%utun0/32      fe80::af27:39e9:de17:1645%utun0  UmCI      utun0
ff02::%utun1/32      fe80::5616:4fc4:7eb1:f6c%utun1  UmCI      utun1
ff02::%utun2/32      fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff02::%utun3/32      fe80::f377:9484:e35a:ca86%utun3  UmCI      utun3
ff02::%utun4/32      fe80::c682:e93b:7892:55c3%utun4  UmCI      utun4
ff02::%utun5/32      fe80::8853:5a51:cfcc:f8c9%utun5  UmCI      utun5
ff02::%utun6/32      fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ridleylherisse@Ridleys-MacBook-Pro ~ % ping 10.0.0.1

PING 10.0.0.1 (10.0.0.1): 56 data bytes
64 bytes from 10.0.0.1: icmp_seq=0 ttl=64 time=14.877 ms
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=12.208 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=12.662 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=10.981 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=20.231 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=6.645 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=14.568 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=14.717 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=9.264 ms
^C
--- 10.0.0.1 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.645/12.897/20.231/3.661 ms
ridleylherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The workstation successfully communicated with the default gateway, confirming that local network connectivity was functioning properly.

4. Step 003 — Verify External Network Connectivity

Command Used

ping 8.8.8.8

Purpose

Used to confirm that traffic could successfully travel beyond the local network.

Evidence

```
ff00::/8          fe80::5616:4fc47eb:f6c%utun1  UmCI      utun1
ff00::/8          fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff00::/8          fe80::f377:9484:e35a:ca8a%utun3  UmCI      utun3
ff00::/8          fe80::c082:e93b:7892:55c%utun4  UmCI      utun4
ff00::/8          fe80::8853:5a51:cfcc:f8c%utun5  UmCI      utun5
ff00::/8          fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ff01::%lo8/32    ::1                          UmCI      lo8
ff01::%np18/32   link#4                      UmCI      np18
ff01::%np11/32   link#5                      UmCI      np11
ff01::%np1/32    link#11                     UmCI      np1
ff01::%en8/32    link#12                     UmCI      en8
ff01::%utun8/32  fe80::af27:39e9:de17:1645%utun0  UmCI      utun8
ff01::%utun1/32  fe80::5616:4fc47eb:f6c%utun1  UmCI      utun1
ff01::%utun2/32  fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff01::%utun3/32  fe80::f377:9484:e35a:ca8a%utun3  UmCI      utun3
ff01::%utun4/32  fe80::c082:e93b:7892:55c%utun4  UmCI      utun4
ff01::%utun5/32  fe80::8853:5a51:cfcc:f8c%utun5  UmCI      utun5
ff01::%utun6/32  fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ff02::%lo8/32    ::1                          UmCI      lo8
ff02::%np18/32   link#4                      UmCI      np18
ff02::%np11/32   link#5                      UmCI      np11
ff02::%np1/32    link#11                     UmCI      np1
ff02::%en8/32    link#12                     UmCI      en8
ff02::%utun8/32  fe80::af27:39e9:de17:1645%utun0  UmCI      utun8
ff02::%utun1/32  fe80::5616:4fc47eb:f6c%utun1  UmCI      utun1
ff02::%utun2/32  fe80::ce81:b1c:bd2c:69e%utun2  UmCI      utun2
ff02::%utun3/32  fe80::f377:9484:e35a:ca8a%utun3  UmCI      utun3
ff02::%utun4/32  fe80::c082:e93b:7892:55c%utun4  UmCI      utun4
ff02::%utun5/32  fe80::8853:5a51:cfcc:f8c%utun5  UmCI      utun5
ff02::%utun6/32  fe80::af27:8863:c81:acbb%utun6  UmCI      utun6
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 10.0.0.1

PING 10.0.0.1 (10.0.0.1): 56 data bytes
64 bytes from 10.0.0.1: icmp_seq=0 ttl=64 time=14.877 ms
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=12.288 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=12.462 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=10.901 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=20.231 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=9.645 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=14.568 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=14.717 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=9.264 ms
^C
--- 10.0.0.1 ping statistics ---
 9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.446/12.877/20.231/3.661 ms
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=36.848 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.797 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=38.574 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=27.983 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=26.494 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=37.499 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=34.763 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=37.231 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=36.157 ms
^C
--- 8.8.8.8 ping statistics ---
 9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 26.494/34.474/38.574/4.070 ms
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

Successful responses confirmed that the device was able to communicate with an external network resource through the configured gateway.

5. Step 004 — Analyze Network Path

Command Used

tracert google.com

Purpose

Used to analyze the path network traffic takes after leaving the local network and identify potential routing issues.

Evidence

```
ff02::Nutun0/32      fe80::af27:39e9:dc17:1645Nutun0 UmcI      utun0
ff02::Nutun1/32      fe80::5616:4fc:47eb:1f6cNutun1 UmcI      utun1
ff02::Nutun2/32      fe80::ce81:b1c:bd2c:69eNutun2 UmcI      utun2
ff02::Nutun3/32      fe80::f377:9484:e35a:ca86Nutun3 UmcI      utun3
ff02::Nutun4/32      fe80::e682:e93b:7892:55c5Nutun4 UmcI      utun4
ff02::Nutun5/32      fe80::8853:5a51:cfc:f8c9Nutun5 UmcI      utun5
ff02::Nutun6/32      fe80::af27:8863:c81:acbbNutun6 UmcI      utun6
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 10.0.0.1

PING 10.0.0.1 (10.0.0.1): 56 data bytes
64 bytes from 10.0.0.1: icmp_seq=0 ttl=64 time=14.877 ms
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=12.288 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=12.662 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=10.901 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=20.231 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=6.645 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=14.568 ms
64 bytes from 10.0.0.1: icmp_seq=7 ttl=64 time=16.717 ms
64 bytes from 10.0.0.1: icmp_seq=8 ttl=64 time=9.264 ms
^C
--- 10.0.0.1 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.645/12.897/20.231/3.661 ms
ridleyherisse@Ridleys-MacBook-Pro ~ % ping 8.8.8.8

PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=114 time=36.848 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=34.797 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=38.574 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=27.983 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=26.494 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=37.499 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=34.763 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=37.231 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=114 time=36.157 ms
^C
--- 8.8.8.8 ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 26.494/34.474/38.574/4.878 ms
ridleyherisse@Ridleys-MacBook-Pro ~ % traceroute google.com

traceroute to google.com (142.251.210.46), 64 hops max, 52 byte packets
 1 10.0.0.1 (10.0.0.1)  7.715 ms  6.468 ms  5.769 ms
 2 172.27.133.226 (172.27.133.226)  21.665 ms
   172.27.133.227 (172.27.133.227)  28.420 ms  30.068 ms
 3 po-64-rur201.plymouth.ma.boston.comcast.net (68.85.186.189)  32.942 ms
 po-54-rur202.plymouth.ma.boston.comcast.net (68.85.186.193)  17.303 ms
 po-54-rur201.plymouth.ma.boston.comcast.net (68.85.186.189)  27.009 ms
 4 po-2-rur201.plymouth.ma.boston.comcast.net (162.151.148.161)  22.697 ms
 po-200-var01.plymouth.ma.boston.comcast.net (162.151.151.85)  40.637 ms
 po-2-rur201.plymouth.ma.boston.comcast.net (162.151.148.161)  28.492 ms
 5 be-384-trscl1.neadham.ma.boston.comcast.net (162.151.151.41)  26.419 ms  24.812 ms
 po-200-var01.plymouth.ma.boston.comcast.net (162.151.151.85)  19.101 ms
 6 be-32041-cs04.newyork.ny.ibone.comcast.net (96.110.42.13)  32.274 ms
 be-304-trscl1.neadham.ma.boston.comcast.net (162.151.151.41)  25.296 ms
 be-32041-cs04.newyork.ny.ibone.comcast.net (96.110.42.13)  27.620 ms
 7 be-32031-cs03.newyork.ny.ibone.comcast.net (96.110.42.9)  32.953 ms
 be-3112-pe12.111eighthave.ny.ibone.comcast.net (96.110.34.34)  21.787 ms
 be-32031-cs03.newyork.ny.ibone.comcast.net (96.110.42.9)  31.409 ms
 8 * be-3312-pe12.111eighthave.ny.ibone.comcast.net (96.110.34.42)  35.688 ms *
 9 * 1c1gaa-ba-in-f14.1e100.net (142.251.210.46)  42.135 ms
   50.219.127.226 (50.219.127.226)  30.123 ms
ridleyherisse@Ridleys-MacBook-Pro ~ %
```

Findings

The traceroute successfully mapped the network path from the workstation to Google. The first hop identified the default gateway (10.0.0.1), followed by multiple ISP routing points before reaching Google's network. This confirmed that the device was able to successfully route traffic beyond the local network.

6. Conclusion

The troubleshooting process verified default gateway configuration, local network communication, routing behavior, and external connectivity. The device successfully communicated with the default gateway and external resources, confirming proper routing functionality.

7. Skills Demonstrated

- Default Gateway Troubleshooting
- Routing Fundamentals
- TCP/IP Concepts
- Network Diagnostics
- Command Line Networking Tools